

PIPE SUPPORT

# ACE Support Plan Edit

Pipe-support plan 300-series — S\_CENTER plate handling, dimension split, ATEXT mirror, EL/FL/GL elevation conversion and BOM.

# Key Purpose

## Why this program

### 1. Auto-finish raw support drawings in one click

– Runs the 300-308 pipeline in memory — layer assign, center lines, symbol edit, dimension create, text editing and BOM — then saves \_ALL.DXF

→ **Effect:** Hours of manual CAD cleanup per support sheet collapse into a single run, with no step forgotten

### 2. One consistent drawing standard for every sheet

– Standard 5-layer reclassification, unified text font and width, DIM\_PIPE dimension style, and Title Block template merge

→ **Effect:** Every support drawing looks identical no matter who modelled it — no hand-fixing layers, fonts or dimension styles

### 3. Correct elevation and annotation without re-drafting

– BOP/COP and EL/FL/GL elevation conversion, S\_CENTER plate handling, dimension split and ATEXT mirror, auto-detected from the drawing

→ **Effect:** Elevation call-outs and dimensions come out clean and right the first time, cutting rework and

**Required inputs :** Support DXF drawing(s) exported from E3D + a Support Title Block DXF template

rebuilds the BOM into the template columns

# Main Window – Item Guide

The screenshot shows the 'Piping Support Drawing Option' dialog box. It has three tabs: 'General', 'Annot', and 'Dimension'. The 'Annot' tab is selected and highlighted with a red circle 1. Inside the 'Annot' tab, there are several options: 'Elevation Expression' with radio buttons for 'BOP' and 'COP' (COP is selected and circled with a red circle 2); 'EL or FL or GL' with radio buttons for 'EL.', 'FL.', and 'GL.' (FL. is selected and circled with a red circle 3); 'Support Output Dwg Scale' with radio buttons for '1:10 Fix Scale Dwg' (selected and circled with a red circle 4) and 'E3D Out Scale Dwg'; 'All Dwg Text Font' with a dropdown menu set to 'Standard' (circled with a red circle 5); 'Support Title Block' with a text field containing 'C:\TEMP\dxfout\WS' and a file explorer icon (circled with a red circle 6); and 'Location Plan Column' with a dropdown menu set to 'North(V):East' (circled with a red circle 7). At the bottom of the dialog, there are four buttons: 'Apply' (circled with a red circle 8), 'OK' (circled with a red circle 9), 'Cancel' (circled with a red circle 10), and 'Help' (circled with a red circle 11).

1. Tabs: General / Annotation / Dimension
2. Elevation Expression (BOP / COP)
3. EL. / FL. / GL. elevation prefix
4. Support Output Dwg Scale (Fix / E3D Out)
5. All Dwg Text Font (Standard / Arial)
6. Support Title Block DXF (click to pick)
7. Location Plan Column mapping
8. Apply (run pipeline)
9. Save options (.opt)
10. Load options (.opt)
11. Cancel

# Processing Steps

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## What 'Apply' runs, in order

1. Assign E3D standard layers
2. Create center lines and scale drawing
3. Convert elevation call-outs (BOP/COP, EL/FL/GL)
4. Edit support symbols
5. Delete line-number text
6. Create dimensions
7. Optional: auto-edit dimensions (Dim Auto Edit)
8. Edit annotation text, circles and leaders
9. Rebuild BOM into template columns
10. Unify all drawing text font and width
11. Reclassify to 5 standard layers (Support Layer Apply)
12. Scale back to paper size
13. Merge Support Title Block and fill fields
14. Save original name + \_ALL.DXF

# Overview

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## What it does

1. Pipe-support plan 300-series — S\_CENTER plate handling
2. Dimension split
3. ATEXT mirror
4. EL/FL/GL elevation conversion and BOM

**Category:** Pipe Support

**Folder:** SUPPORT\_DWG\_EDIT/ACE\_SUPP\_E3D

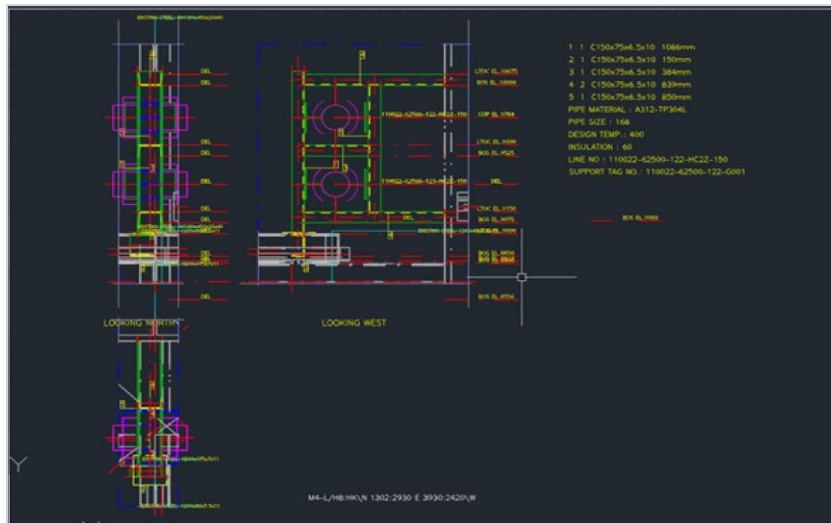
**Main scripts:** ACE\_SUPP\_RUNALL.PY

**Scripts / Status:** 13 scripts · Operational

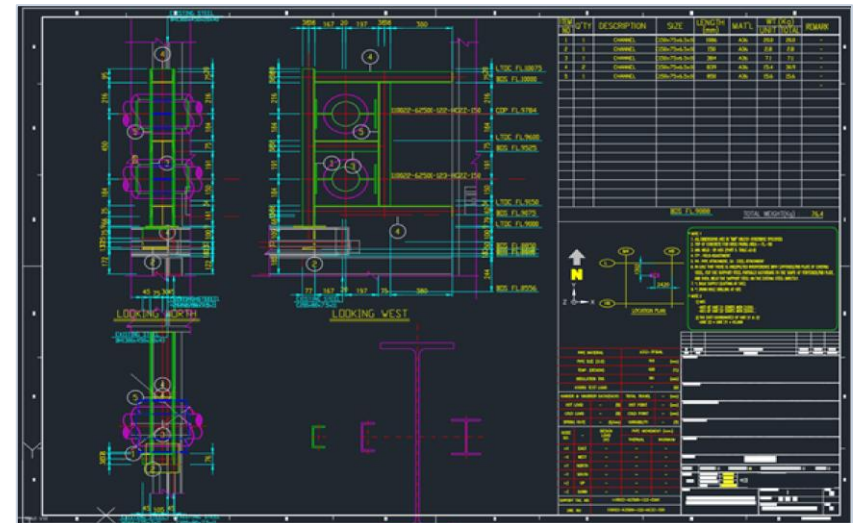
**Tags:** Support · S\_CENTER · ATEXT · BOM

# Output Samples

## Output Samples (1/3)



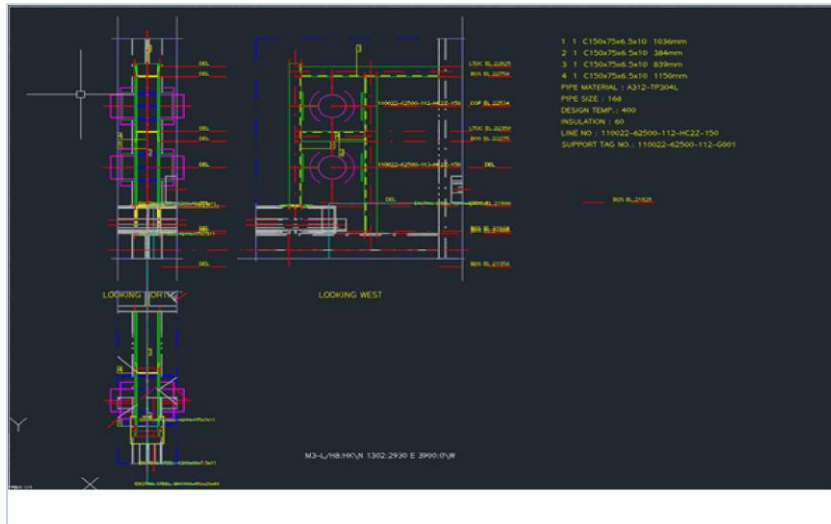
Support Plan Input 1



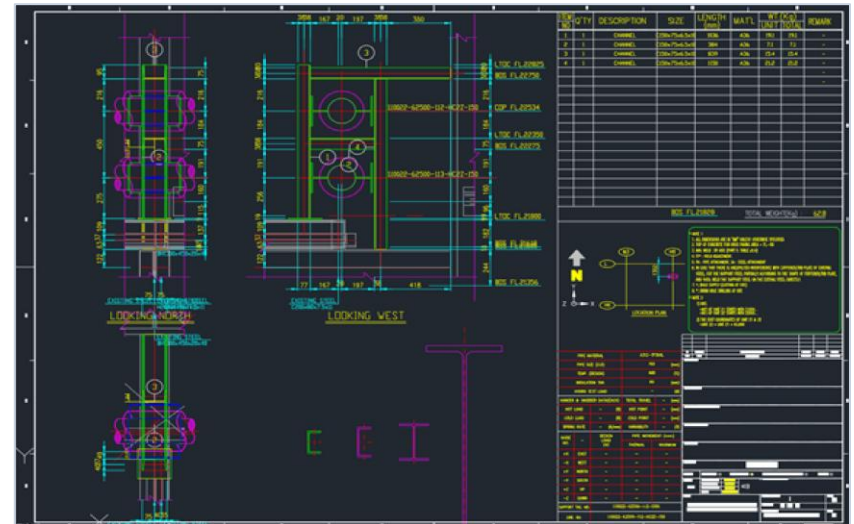
Support Plan Auto Output 1

# Output Samples

## Output Samples (2/3)



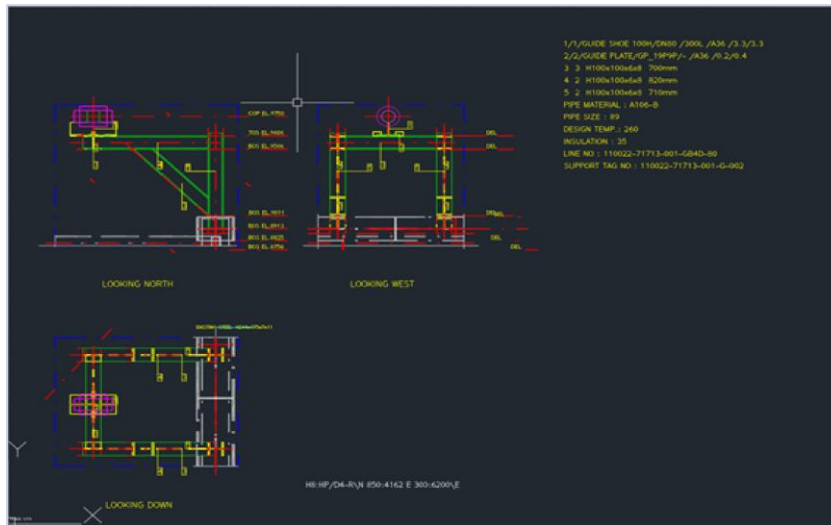
Support Plan Input 2



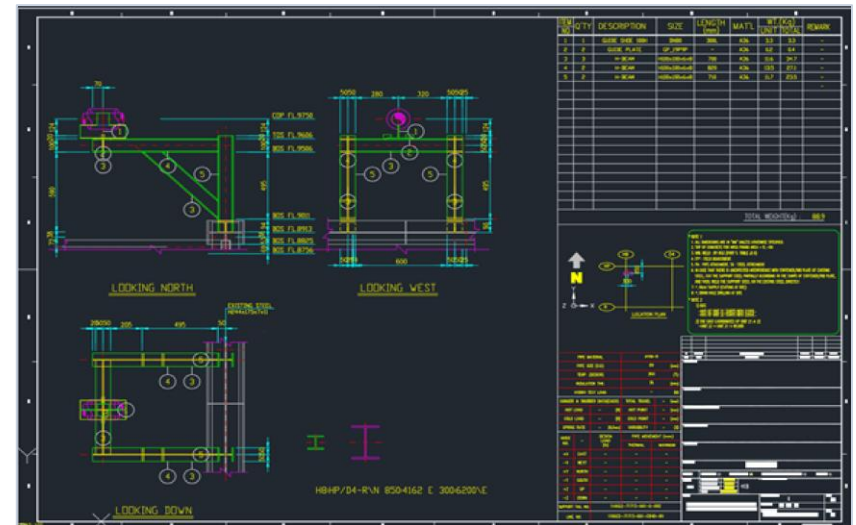
Support Plan Auto Output 2

# Output Samples

## Output Samples (3/3)



Support Plan Input 3



Support Plan Auto Output 3

# Notes & Information

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- Part of the in-house Plant Engineering automation suite — category: Pipe Support.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: ACE\_SUPP\_RUNALL.PY.
- Output samples and the program window above reflect the current build.

# Thank You

ACE Support Plan Edit · Plant Engineering Total Service System